

POSITIVE ESTIMATION OF THE BETWEEN-STUDY VARIANCE IN META-ANALYSIS

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Summary: Positive estimators of the between-study variance are proposed. Explicit variance formulae for the estimators are given and approximate confidence intervals for the between-group variance are constructed, thus helping to partly solve a long outstanding and open problem. By Monte Carlo simulation, the bias and standard deviation of the proposed estimators are compared with a standard estimator in meta-analysis. Attained confidence coefficients of the constructed confidence intervals are also presented. To illustrate the application of the proposed procedures, three examples are given, the first is from agriculture, the other two are from clinical trials with normal and binary outcomes, respectively.

1. Introduction

In the usual random effects ANOVA model, the problem of variance component estimation is widely documented. The traditional one-way ANOVA estimator of the between-group variance component, also called the between-study variance or the heterogeneity variance in meta-analysis,

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is unbiased but can assume negative values, (Thompson, 1962 and Wang, 1967). In practice, whenever unbiased estimators become negative, they are truncated at zero, thus making them biased and have now a positive probability to take on the value zero which, in most applications, is not a very useful value and contradicts practical understanding. Such estimators also have a further disadvantage that in concrete situations their estimates of precision cannot be explicitly computed, which means that confidence bounds on the variance components cannot be constructed. In meta-analysis, the same problem is encountered by using, for example, the DerSimonian and Laird (1986) estimator, which of late may be regarded as the standard estimator of the between-study variance. In fact Hardy and Thompson (1996) observe that since the DerSimonian-Laird estimator is truncated, there is no possibility of obtaining confidence intervals on the between-study variance.

Pinheiro and Bates (1996) describe five parameterizations for variance-covariance matrices that ensure positive semi-definiteness. Apart from this approach being iterative in nature, it lacks a clearly spelled out guide to discriminate among the options. Other possible procedures for obtaining variance component estimates include the maximum likelihood (ML) and the restricted maximum likelihood (REML). These two procedures use numerical iteration and their problems are well documented in Searle et al. (1992, p. 242–243, 290–292). Usually there is the problem of convergence, and if it is achieved, whether it is global or local convergence. In general, likelihood methods may require knowledge on numerical analysis of which many practitioners in, for example, the Biomedical sciences are ill equipped. Therefore, a more direct and straightforward procedure is required which should also be computationally more friendly and easy to implement in

any statistical package. Further let us remark, that Hartung (1981) gives a sequential (penalty) algorithm converging to the positive semi-definite minimum biased quadratic estimation function for a nonnegative variance component which, however, needs some numerical effort too.

So in this paper, we propose two finite and easily computable positive (almost everywhere) estimators of the between-study variance in a one-way random effects model for meta-analysis. In section 2, we present our working model together with the moments estimator of the between-study variance by DerSimonian and Laird (1986). In section 3, we derive two positive estimators of the between-study variance and give their explicit variance formulae. Confidence limits on the between-study variance are also presented in section 3. Section 4 presents simulation results on the biases and standard deviations of the two proposed positive estimators and the truncated version of the DerSimonian-Laird estimator. Reported also, see section 4, are results on the attained confidence coefficients of the constructed confidence bounds. Then we illustrate how the procedures developed can be adapted to two-armed trials with normally distributed responses, see section 5, and to data with binary outcomes, using the risk difference as the outcome measure, see section 6. Note that extension to other metrics for comparing proportions, for example the log odds ratio and the log relative risk, can easily be done using similar arguments as those presented here for the risk difference. Three examples are given in section 7, the first one is from agriculture on research in artificial insemination of cows using semen from several bulls, and the other two are on combining data from controlled clinical trials, one with normally distributed responses and one with binary outcomes.

2. Model

Consider a situation where K studies are available, with the i th study having n_i observations, $i = 1, \dots, K$, which follow the model

$$y_i = \mu_i + e_i = \mu + a_i + e_i, \quad i = 1, \dots, K > 2, \quad (1)$$

see for example, Cochran (1954). The variables $a_1, \dots, a_K, e_1, \dots, e_K$ are mutually independent and normally distributed with $a_i \sim (0, \sigma_a^2)$, $e_i \sim (0, \sigma_i^2)$, $\sigma_a^2 \geq 0$, $\sigma_i^2 > 0$, and μ is the overall treatment difference common to all studies. In a meta-analysis, there is usually available from the i th study an estimate of the outcome measure of interest, $\hat{\mu}_i$, together with an estimate of its variance, $\hat{\sigma}_i^2 = \xi_i$, on ν_i degrees of freedom, $i = 1, \dots, K$. These estimates are such that $y_i = \hat{\mu}_i \sim N(\mu, \tau_i^2)$, $\tau_i^2 = \sigma_a^2 + \sigma_i^2$.

Let us now define $\omega_\Sigma = \sum_{i=1}^K \omega_i$, $\omega_i = 1/\xi_i$, $i = 1, \dots, K$. For $\text{var}(\xi_i) = 2 \cdot \sigma_i^4/\nu_i$, we can take $\widehat{\text{var}}(\xi_i) = 2 \cdot \xi_i^2/\nu_i$ as its estimate or the best invariant unbiased estimator $\widehat{\text{var}}(\xi_i) = 2 \cdot \xi_i^2/(\nu_i + 2)$ of Hartung and Voet (1986). The standard estimate of σ_a^2 in meta-analysis is a moment estimator suggested by DerSimonian and Laird (1986), namely,

$$\hat{\sigma}_{a,DL}^2 = \max \left\{ 0, \hat{\sigma}_{a,1}^2 \right\},$$

where

$$\hat{\sigma}_{a,1}^2 = \frac{\omega_\Sigma}{\omega_\Sigma^2 - \sum_{i=1}^K \omega_i^2} \left\{ \sum_{i=1}^K \omega_i \left(y_i - \sum_{j=1}^K \frac{\omega_j}{\omega_\Sigma} y_j \right)^2 - K + 1 \right\}, \quad (2)$$

see also Mengersen, Tweedie and Biggerstaff (1995), Biggerstaff and Tweedie (1997). The estimator $\hat{\sigma}_{a,DL}^2$ has been widely used, especially to draw inference on the overall treatment effect parameter, refer to, for instance, Cochran (1954), DerSimonian and Laird (1986) and Hartung and Makambi (2001). The terms $c_i = \omega_i/\omega_\Sigma$ are named as the inverse variance weights.

3. Proposed positive estimators and confidence bounds on σ_a^2

Now, let $b = (b_1, \dots, b_K)'$ be weights such that $\sum_{i=1}^K b_i = 1$, $b_i < 1/2$, $i = 1, \dots, K$. Define an unbiased estimate of μ by $\hat{\mu}(b) = \sum_{i=1}^K b_i \hat{\mu}_i$ with $\text{var}\{\hat{\mu}(b)\} = \sum_{i=1}^K b_i^2 \tau_i^2$; and a quadratic form $Q(b)$ by $Q(b) = \sum_{i=1}^K \gamma_i \{\hat{\mu}_i - \hat{\mu}(b)\}^2$, where $\gamma_i = b_i^2 / \left\{ (1 - 2b_i) \cdot \sum_{j=1}^K b_j (1 - b_j) / (1 - 2b_j) \right\}$.

Lemma 1

The estimator

$$\hat{\sigma}_{a,2}^2 = \frac{1}{\sum_{j=1}^K b_j^2} \cdot \left\{ Q(b) - \sum_{i=1}^K b_i^2 \xi_i \right\} \quad (3)$$

is unbiased for σ_a^2 .

Proof

Using Rao et al. (1981),

$$\begin{aligned} E\{Q(b)\} &= \sum_{i=1}^K \gamma_i \left\{ (1 - 2b_i) \tau_i^2 + \sum_{i=1}^K b_i^2 \tau_i^2 \right\} \\ &= \frac{1 + \sum_{i=1}^K b_i^2 / (1 - 2b_i)}{\sum_{i=1}^K b_i (1 - b_i) / (1 - 2b_i)} \cdot \sum_{i=1}^K b_i^2 \tau_i^2 \\ &= \sum_{i=1}^K b_i^2 \tau_i^2. \end{aligned} \quad (4)$$

Therefore,

$$\begin{aligned} E(\hat{\sigma}_{a,2}^2) &= \frac{1}{\sum_{j=1}^K b_j^2} \cdot E\left\{ Q(b) - \sum_{i=1}^K b_i^2 \xi_i \right\} \\ &= \sigma_a^2. \end{aligned}$$

Now, $E(\hat{\sigma}_{a,2}^2) = [E\{Q(b)\} - E(\sum_{i=1}^K b_i^2 \xi_i)] / \sum_{j=1}^K b_j^2$, implying that

$$E\left\{ \frac{Q(b)}{\sum_{j=1}^K b_j^2} \right\} = E(\hat{\sigma}_{a,2}^2) + E\left(\frac{\sum_{i=1}^K b_i^2 \xi_i}{\sum_{j=1}^K b_j^2} \right) > E\left(\frac{\sum_{i=1}^K b_i^2 \xi_i}{\sum_{j=1}^K b_j^2} \right). \quad (5)$$

However, after a realization from a random experiment, $E\{Q(b)\}$ and $E\left(\sum_{i=1}^K b_i^2 \xi_i\right)$ are, respectively, replaced by, say, $Q(b)$ and $\sum_{i=1}^K b_i^2 \xi_i$. This may lead to $Q(b) < \sum_{i=1}^K b_i^2 \xi_i$, meaning that the estimate of σ_a^2 is negative. This is a problem which restricts the use of most unbiased estimators of variance components. Usually solutions are given in the name of truncated estimators. By using truncated estimators, apart from losing the desirable property of unbiasedness, in many applications, it sometimes becomes difficult to accept zero as an estimate of the between-study variance when it is well known that there is variation among the groups under consideration. For example, in genetics we know that genetic variation exists among, say, bulls whose sperms are used for insemination. This and many other similar situations beg for a positive estimate of the between-group variance. Instead of using complicated numerical algorithms as, for instance, in Hartung (1981, sec. 4), we will proceed as follows:

Interpret $Q_1(b) = Q(b) / \sum_{j=1}^K b_j^2$ as a positive estimate of σ_a^2 which requires some correction and define an estimator of σ_a^2 by $\hat{\sigma}_a^2(b) = \delta \cdot Q_1(b)$, $\delta > 0$. For $r_i = b_i^2 / \sum_{j=1}^K b_j^2$, the magnitude of the bias of $\hat{\sigma}_a^2(b)$ is given by

$$\begin{aligned}
 |Bias\{\hat{\sigma}_a^2(b)\}| &= |E\{\delta \cdot Q_1(b)\} - \sigma_a^2| \\
 &= \left| \delta \cdot \sigma_a^2 + \delta \cdot \sum_{i=1}^K r_i \sigma_i^2 - \sigma_a^2 \right| \\
 &= \left\| \begin{pmatrix} \delta - 1 \\ \delta \end{pmatrix} \cdot \begin{pmatrix} \sigma_a^2 \\ \sum_{i=1}^K r_i \sigma_i^2 \end{pmatrix} \right\| \\
 &\leq \left\| \begin{pmatrix} \delta - 1 \\ \delta \end{pmatrix} \right\| \cdot \left\| \begin{pmatrix} \sigma_a^2 \\ \sum_{i=1}^K r_i \sigma_i^2 \end{pmatrix} \right\| \quad (6)
 \end{aligned}$$

by the Cauchy-Schwarz inequality, where $\|(\cdot)\|$ represents the Euclidean norm of $(\cdot)$. According to the uniformly minimum bias principle by Hartung (1981), we have to minimize $(\delta - 1)^2 + \delta^2$ for $\delta > 0$, giving $\delta = \frac{1}{2}$.

Now, to adjust for bias, let $\hat{\sigma}_a^2(\eta) = \eta \cdot \frac{1}{2} \cdot Q_1(b) = \eta \cdot \hat{\sigma}_a^2(b)$ such that

$$E \left\{ \eta \cdot \hat{\sigma}_a^2(b) + \eta \cdot \sum_{i=1}^K r_i \xi_i \right\} = E \{ Q_1(b) \}, \quad (7)$$

giving

$$\eta = \frac{E \{ Q_1(b) \}}{(1/2) \cdot E \{ Q_1(b) \} + \sum_{i=1}^K r_i \sigma_i^2},$$

which can be estimated by

$$\eta^* = \frac{2 \cdot Q_1(b)}{Q_1(b) + 2 \cdot \sum_{i=1}^K r_i \xi_i}$$

and henceforth regarded as fixed. The above arguments lead us to a positive estimator of σ_a^2 which is given by

$$\hat{\sigma}_a^2(\eta) = \eta^* \cdot \frac{1}{2} \cdot Q_1(b) = \frac{Q_1(b)}{Q_1(b) + 2 \cdot \sum_{i=1}^K r_i \xi_i} \cdot Q_1(b). \quad (8)$$

Lemma 2

The variance of $\hat{\sigma}_a^2(\eta)$ is given by

$$\begin{aligned} \text{var} \left\{ \hat{\sigma}_a^2(\eta) \right\} &= \frac{\lambda_1^2}{\left(\sum_{j=1}^K b_j^2 \right)^2} \cdot \text{var} \{ Q(b) \} \\ &= \frac{\lambda_1^2}{\left(\sum_{j=1}^K b_j^2 \right)^2} \cdot \left[\sum_{i=1}^K \gamma_i^2 \{ (1 - 2b_i) \tau_i^2 + \text{var}(\hat{\mu}(b)) \}^2 \right. \\ &\quad \left. + \sum_{i=1}^K \sum_{j \neq i=1}^K \gamma_i \gamma_j \{ \text{var}(\hat{\mu}(b)) - b_i^2 \tau_i^2 - b_j^2 \tau_j^2 \}^2 \right], \quad (9) \end{aligned}$$

for $\lambda_1 = Q_1(b) / \left\{ Q_1(b) + 2 \cdot \sum_{i=1}^K r_i \xi_i \right\}$.

The proof of the above lemma is straightforward by using equation (30) of Rao et al. (1981).

We see that the proposed estimator of σ_a^2 , apart from being finite and positive, has an explicit variance formula which can be used, for example, in the construction of confidence intervals.

Consider once again the quadratic form $Q(b)$ and define $\nu_Q = 2 \cdot [E\{Q(b)\}]^2 / \text{var}\{Q(b)\}$. By Patnaik's (1949) approximation, the variable $\nu_Q \cdot Q(b) / E\{Q(b)\}$ is distributed approximately as a $\chi_{\nu_Q}^2$. From equation (4) above,

$$E\{Q(b)\} = \sum_{i=1}^K b_i^2 \tau_i^2 = \sigma_a^2 \cdot \sum_{i=1}^K b_i^2 + \sum_{i=1}^K b_i^2 \sigma_i^2,$$

so that the approximate probability statement

$$P \left[\chi_{\nu_Q; \kappa/2}^2 \leq \nu_Q \cdot \frac{Q(b)}{E\{Q(b)\}} \leq \chi_{\nu_Q; 1-\kappa/2}^2 \right] \approx 1 - \kappa \quad (10)$$

results in an approximate $(1 - \kappa)$ 100%-confidence interval on σ_a^2 which is given by

$$CI_1(\sigma_a^2) : \left[\frac{\hat{\nu}_Q \cdot Q_1(b)}{\chi_{\hat{\nu}_Q; 1-\kappa/2}^2} - \sum_{i=1}^K r_i \xi_i, \frac{\hat{\nu}_Q \cdot Q_1(b)}{\chi_{\hat{\nu}_Q; \kappa/2}^2} - \sum_{i=1}^K r_i \xi_i \right], \quad (11)$$

where $\chi_{\hat{\nu}_Q; \kappa/2}^2$ and $\chi_{\hat{\nu}_Q; 1-\kappa/2}^2$ are, respectively, the lower and upper points of $\chi_{\hat{\nu}_Q}^2$, and $\hat{\nu}_Q = 2 \cdot Q^2(b) / \widehat{\text{var}}\{Q(b)\}$; with $\widehat{\text{var}}\{Q(b)\}$ obtained by replacing τ_i^2 by $\hat{\tau}_i^2 = \hat{\sigma}_a^2(\eta) + \xi_i$, $i = 1, \dots, K$. Note that if the limits take on negative values, they are truncated at zero.

Remark 1

The condition that $b_i < 1/2$, $i = 1, \dots, K$, helps to avoid the tendency of one of the studies dominating the estimation process, an observation already made by Cochran (1954).

Corollary 1

Let $\beta_i = w_i/w_\Sigma$, $w_\Sigma = \sum_{i=1}^K w_i$, $w_i = 1/\sigma_i^2$, see also $\hat{\sigma}_{a,1}^2$ in equation (2) above. Another positive estimator of σ_a^2 is given by

$$\hat{\sigma}_a^2(\lambda_2) = \frac{\lambda_2}{1 - \sum_{i=1}^K \beta_i^2} \cdot \sum_{i=1}^K \beta_i \left(y_i - \sum_{j=1}^K \beta_j y_j \right)^2, \quad (12)$$

where

$$\lambda_2 = \frac{\sum_{i=1}^K w_i \left(y_i - \sum_{j=1}^K b_j^* y_j \right)^2}{2(K-1) + \sum_{i=1}^K w_i \left(y_i - \sum_{j=1}^K b_j^* y_j \right)^2}, \quad b_i^* = \frac{w_i}{w_\Sigma}.$$

This estimator is obtained by using arguments similar to those that lead to $\hat{\sigma}_a^2(\eta)$ above. In the following, λ_2 is considered as fixed.

Corollary 2

The variance of $\hat{\sigma}_a^2(\lambda_2)$ is given by

$$\begin{aligned} \text{var} \left\{ \hat{\sigma}_a^2(\lambda_2) \right\} = & \frac{\lambda_2^2}{\left(1 - \sum_{i=1}^K \beta_i^2 \right)^2} \cdot \left[\sum_{i=1}^K \beta_i^2 \left\{ (1-2\beta_i) \tau_i^2 + \text{var}(\hat{\mu}(\beta)) \right\}^2 \right. \\ & \left. + \sum_{i=1}^K \sum_{j \neq i=1}^K \beta_i \beta_j \left(\text{var}(\hat{\mu}(\beta)) - \beta_i^2 \tau_i^2 - \beta_j^2 \tau_j^2 \right)^2 \right]. \quad (13) \end{aligned}$$

This corollary follows from considerations similar to those of lemma 2 above.

If we now set $Q(\beta) = \sum_{i=1}^K \beta_i \left(y_i - \sum_{j=1}^K \beta_j y_j \right)^2$, then it can be shown that

$$E \{ Q(\beta) \} = \sigma_a^2 \cdot \left(1 - \sum_{i=1}^K \beta_i^2 \right) + \sum_{i=1}^K (\beta_i - \beta_i^2) \cdot \sigma_i^2.$$

Therefore, an approximate $(1 - \kappa)100\%$ -confidence interval on σ_a^2 can be constructed along the lines of $CI_1(\sigma_a^2)$. That is, for $r_i^* = (b_i^* - b_i^{*2}) / (1 - \sum_{i=1}^K b_i^{*2})$,

$$CI_2(\sigma_a^2) : \left[\frac{\widehat{\nu}_{Q^*} \cdot Q(b^*)}{\left(1 - \sum_{i=1}^K b_i^{*2}\right) \cdot \chi_{\widehat{\nu}_{Q^*}, 1-\kappa/2}^2} - \sum_{i=1}^K r_i^* \xi_i, \right. \\ \left. \frac{\widehat{\nu}_{Q^*} \cdot Q(b^*)}{\left(1 - \sum_{i=1}^K b_i^{*2}\right) \cdot \chi_{\widehat{\nu}_{Q^*}, \kappa/2}^2} - \sum_{i=1}^K r_i^* \xi_i \right], \quad (14)$$

where $Q(b^*) = \sum_{i=1}^K b_i^* \left(y_i - \sum_{j=1}^K b_j^* y_j\right)^2$ and $\widehat{\nu}_{Q^*} = 2 \cdot Q^2(b^*) / \widehat{var}\{Q(\beta)\}$, and $\widehat{var}\{Q(\beta)\}$ is obtained by replacing τ_i^2 by $\widehat{\tau}_i^2 = \widehat{\sigma}_a^2(\lambda_2) + \xi_i$, $i = 1, \dots, K$. Here also, the limits are truncated at zero if they take on negative values.

4. Simulation results and discussion

A simulation study, with patterns shown in Table I below, was used to judge the performance of the proposed positive estimators of the between-study variance with respect to their bias and standard deviation. For comparison we have also included the corresponding results of the bias and standard deviation of the DerSimonian-Laird estimator (see Table II and III). For groups of size $K=3$, Table II, and $K=6$, Table III, 10,000 runs were made for each pattern in Table I with the between-group variances set at $\sigma_a^2=0.0, 0.05, 0.5, 1.0, 5.0$ and 10.0 .

For the choice of weights b_i , $i = 1, \dots, K$, to avoid domination of a single study (in line with remark 1 above), we used the following procedure: Let $c_i = \omega_i / \omega_{\Sigma} > 0$ be a positive estimate of b_i , $i = 1, \dots, K$, and φ a constant such that $0 < \varphi < \frac{1}{2} - \frac{1}{K}$, then we set

$$b_i = \begin{cases} c_i, & \text{if } c_i \leq \frac{1}{2} - \varphi, \quad i = 1, \dots, K, \\ \frac{1}{2} - \varphi, & \text{for } i = i_0 \text{ with } c_{i_0} = \max \{c_j : c_j > \frac{1}{2} - \varphi\}, \\ \left(\frac{1}{2} + \varphi\right) \cdot c_i / \sum_{i_0 \neq i=1}^K c_i & i_0 \neq i = 1, \dots, K. \end{cases} \quad (15)$$

If (15) gives $b_i \leq \frac{1}{2} - \varphi$, $i = 1, \dots, K$, then stop, otherwise, put $\varphi = \varphi/2$ and start (15) once again, etc. In our simulations, we have chosen $\varphi = K^{-3}$.

Table IV reports results on attained confidence coefficients for the confidence bounds $CI_1(\sigma_a^2)$ and $CI_2(\sigma_a^2)$ in section 3 above for $K = 3$ and $K = 6$ with 10,000 runs for each pattern. These results are for a two-sided confidence interval with $\kappa = 0.05$.

Table I. Sample designs for $K = 3, 6$ for the simulation results in Table II, III and IV

	Study i	$K = 3$			$K = 6$					
		1	2	3	1	2	3	4	5	6
A1	n_i	20	20	20	20	20	20	20	20	20
	$\sigma_{e_i}^2$	4	4	4	4	4	4	4	4	4
A2	n_i	20	20	20	20	20	20	20	20	20
	$\sigma_{e_i}^2$	1	3	5	1	3	5	1	3	5
B1	n_i	10	20	30	10	20	30	10	20	30
	$\sigma_{e_i}^2$	4	4	4	4	4	4	4	4	4
B2	n_i	10	20	30	10	20	30	10	20	30
	$\sigma_{e_i}^2$	1	3	5	1	3	5	1	3	5
B3	n_i	10	20	30	10	20	30	10	20	30
	$\sigma_{e_i}^2$	5	3	1	5	3	1	5	3	1

All three estimators have nonnegative bias for $\sigma_a^2 = 0$ which decreases with increasing K , see Table II and III. For $\sigma_a^2 = 0.05$, the three estimators have nonnegative bias for both $K = 3$ and $K = 6$. In general, $\hat{\sigma}_a^2(\lambda_2)$ seems to underestimate σ_a^2 for $\sigma_a^2 \geq 0.5$. For example, for $K = 3$ and $\sigma_a^2 = 10.0$, the underestimation is between 3% and 5%, and for $K = 6$ the underestimation is between 0.6% and 6%. Also for $\sigma_a^2 \geq 0.5$, the estimators $\hat{\sigma}_a^2(\eta)$ and $\hat{\sigma}_{a,DL}^2$ generally underestimate σ_a^2 for $K = 3$, a trend which is largely reversed for $K = 6$.

Table II. Bias (B) and Standard Deviation (SD) for three between-study variance estimators, $K = 3$.

$K = 3$							
σ_a^2	Plan	$\hat{\sigma}_a^2(\eta)$		$\hat{\sigma}_a^2(\lambda_2)$		$\hat{\sigma}_{a,DL}^2$	
		B	SD	B	SD	B	SD
0.00	A1	0.099	0.17	0.089	0.13	0.074	0.15
	A2	0.053	0.09	0.061	0.09	0.052	0.11
	B1	0.093	0.18	0.096	0.15	0.081	0.17
	B2	0.062	0.11	0.060	0.09	0.051	0.10
	B3	0.061	0.12	0.078	0.13	0.069	0.14
0.05	A1	0.086	0.23	0.076	0.19	0.065	0.22
	A2	0.034	0.14	0.043	0.13	0.039	0.15
	B1	0.082	0.25	0.085	0.21	0.075	0.24
	B2	0.051	0.16	0.050	0.14	0.045	0.16
	B3	0.051	0.18	0.059	0.17	0.054	0.19
0.50	A1	0.031	0.73	0.012	0.64	0.044	0.69
	A2	0.002	0.78	0.000	0.66	0.031	0.69
	B1	0.012	0.76	-0.001	0.64	0.027	0.69
	B2	-0.022	0.73	-0.030	0.61	-0.002	0.64
	B3	-0.014	0.76	-0.014	0.67	0.014	0.70
1.00	A1	-0.086	1.21	-0.056	1.10	0.004	1.16
	A2	0.032	1.47	0.001	1.21	0.053	1.25
	B1	-0.030	1.49	-0.061	1.21	-0.002	1.26
	B2	-0.040	1.13	-0.045	1.10	0.008	1.14
	B3	-0.079	1.41	-0.081	1.23	-0.033	1.27
5.00	A1	0.114	6.53	-0.005	5.65	0.014	5.69
	A2	-0.284	6.20	-0.207	5.26	-0.115	5.28
	B1	-0.286	5.71	-0.256	4.91	-0.120	4.95
	B2	-0.199	6.20	-0.172	5.24	-0.076	5.27
	B3	-0.465	6.40	-0.402	5.57	-0.308	5.60
10.00	A1	-0.119	12.96	-0.364	10.75	-0.215	10.78
	A2	0.259	12.43	0.260	10.33	0.366	10.35
	B1	-0.104	12.94	-0.415	10.46	-0.255	10.49
	B2	-0.037	12.29	0.195	10.83	0.306	10.84
	B3	0.152	14.21	0.250	12.47	0.361	12.49

Table III. Bias (B) and Standard Deviation (SD) of three between-study variance estimators, $K = 6$.

$K = 6$							
σ_a^2	Plan	$\hat{\sigma}_a^2(\eta)$		$\hat{\sigma}_a^2(\lambda_2)$		$\hat{\sigma}_{a,DL}^2$	
		B	SD	B	SD	B	SD
0.00	A1	0.090	0.12	0.082	0.09	0.016	0.10
	A2	0.033	0.05	0.046	0.052	0.008	0.06
	B1	0.089	0.12	0.095	0.10	0.073	0.12
	B2	0.057	0.09	0.053	0.06	0.037	0.07
	B3	0.021	0.03	0.041	0.05	0.030	0.05
0.05	A1	0.068	0.13	0.060	0.11	0.038	0.13
	A2	0.025	0.12	0.031	0.08	0.025	0.10
	B1	0.075	0.17	0.077	0.13	0.060	0.16
	B2	0.043	0.12	0.037	0.08	0.026	0.10
	B3	0.019	0.12	0.032	0.09	0.028	0.11
0.50	A1	0.014	0.50	-0.013	0.41	0.026	0.46
	A2	-0.006	0.65	-0.025	0.42	0.013	0.44
	B1	0.027	0.57	0.001	0.44	0.040	0.48
	B2	0.017	0.59	-0.010	0.42	0.030	0.44
	B3	0.015	0.73	-0.013	0.48	0.022	0.51
1.00	A1	-0.077	0.86	-0.099	0.72	-0.024	0.76
	A2	-0.020	1.25	-0.044	0.81	0.014	0.83
	B1	-0.023	1.01	-0.036	0.82	0.040	0.86
	B2	0.047	1.16	0.012	0.84	0.078	0.86
	B3	-0.016	1.39	-0.034	0.93	0.017	0.95
5.00	A1	-0.134	4.07	-0.138	3.41	0.011	3.43
	A2	0.220	5.79	0.042	3.84	0.131	3.85
	B1	0.107	5.29	-0.035	3.82	0.116	3.84
	B2	0.163	5.12	-0.018	3.52	0.087	3.53
	B3	0.003	6.79	-0.011	4.73	0.067	4.73
10.00	A1	0.010	8.21	0.033	7.11	0.198	7.12
	A2	0.016	12.17	-0.068	7.90	0.026	7.91
	B1	-0.629	8.47	-0.572	6.88	-0.404	6.90
	B2	0.018	9.95	-0.072	6.96	0.040	6.96
	B3	0.259	13.32	0.081	8.62	0.163	8.98

Table IV. Attained confidence coefficients for the confidence intervals of the between-study variance for $K = 3, 6$ and $\kappa = 0.05$

σ_a^2	Plan	Estimates of $(1 - \kappa) 100\%$			
		$K = 3$		$K = 6$	
		$CI_1(\sigma_a^2)$	$CI_2(\sigma_a^2)$	$CI_1(\sigma_a^2)$	$CI_2(\sigma_a^2)$
0.00	A1	94.3	94.1	94.4	94.3
	A2	95.6	94.0	95.9	94.3
	B1	95.0	94.2	94.6	94.5
	B2	94.8	94.0	94.1	94.3
	B3	94.7	94.2	96.2	95.3
0.05	A1	94.9	94.3	94.8	94.6
	A2	96.1	94.7	96.0	94.5
	B1	95.1	94.4	95.4	94.8
	B2	95.1	94.6	95.8	94.3
	B3	95.7	94.0	96.3	94.0
0.50	A1	94.6	94.5	95.6	94.8
	A2	96.0	95.0	96.2	95.4
	B1	96.1	94.7	95.2	94.8
	B2	95.7	94.9	95.6	94.7
	B3	95.9	94.7	96.5	94.9
1.00	A1	95.8	95.1	95.1	94.7
	A2	96.3	94.9	96.3	94.7
	B1	96.3	95.0	95.6	94.8
	B2	96.0	94.5	95.2	94.8
	B3	95.8	94.4	96.6	94.8
5.00	A1	96.0	94.9	95.6	95.2
	A2	96.1	94.8	96.5	95.4
	B1	95.6	95.0	95.8	95.8
	B2	96.4	94.8	96.6	95.4
	B3	96.0	94.8	97.0	95.6
10.00	A1	95.8	94.8	96.1	95.2
	A2	96.3	95.0	97.0	95.8
	B1	95.9	94.4	96.4	95.3
	B2	95.7	94.9	97.4	95.3
	B3	96.1	94.8	96.5	96.0

The standard deviations of the three estimators increase with σ_a^2 and decrease with K . The results in Table IV indicate that for $K = 3$, $CI_1(\sigma_a^2)$ sometimes overstates the nominal confidence coefficient but, in general this interval can be regarded as attaining acceptable confidence coefficients. An almost similar trend is manifested for $K = 6$. Notice that for $K = 6$ in the unbalanced case when smaller sample sizes are paired with larger variances (Plan B3), the interval always attains confidence coefficients of over 96.0%.

For $K = 3$, the interval $CI_2(\sigma_a^2)$ attains confidence coefficients which are always $\leq 95\%$, except plan A1 for $\sigma_a^2 = 1.0$. However, the confidence coefficients are generally close to 95%. For $K = 6$, the interval attains confidence coefficients near the nominal confidence coefficient. On comparing $CI_1(\sigma_a^2)$ and $CI_2(\sigma_a^2)$, it can be said that both intervals attain acceptable confidence coefficients but $CI_1(\sigma_a^2)$ tends towards being more conservative when the value of σ_a^2 becomes large.

5. The difference of means

Let be given $K > 2$ independent experiments each consisting of two independent treatment arms, $k=1, 2$, with normally distributed responses, respectively. In the i th trial, $i = 1, \dots, K$, we observe $n_{ik} \geq 2$ subjects and get the means $y_{ik} = \hat{\mu}_{ik}$ with ξ_{ik} as estimates of their variances, associated to $\nu_{ik} = n_{ik} - 1$ degrees of freedom.

The outcome measure is the expected difference of the means, μ , in the i th study estimated by $y_i = \hat{\mu}_i = y_{i1} - y_{i2}$, for which model (1) is adapted, where now σ_a^2 describes the heterogeneity of the mean differences as treatment effects. The study specific variance estimate of y_i is $\xi_i = \xi_{i1} + \xi_{i2}$, and by Patnaik's approximation the related degrees of freedom can be estimated by $\hat{\nu}_i = \xi_i^2 / \{ \xi_{i1}^2 / (n_{i1} + 1) + \xi_{i2}^2 / (n_{i2} + 1) \}$, see sections 2 and 3.

6. The risk difference approach

We now consider binary outcomes from, say, multicentre studies with two arms. Let z_{ik} be binomially distributed with parameters $n_{ik} \geq 2$ and p_{ik} , $0 < p_{ik} < 1$, with the risk difference $\mu = p_{i1} - p_{i2} = p_1 - p_2$, assumed identical in all the studies and $\hat{p}_{ik} = z_{ik}/n_{ik}$ with $\text{var}(\hat{p}_{ik}) = \sigma_{ik}^2 = p_{ik}(1 - p_{ik})/n_{ik}$, $i = 1, \dots, K$, $k = 1, 2$.

For the difference of the empirical rates with realization: $\hat{p}_{i1}^* - \hat{p}_{i2}^*$, there is allowed that one really observes $\hat{\mu}_i = a_i + \hat{p}_{i1} - \hat{p}_{i2}$, $i = 1, \dots, K$, where the stochastically independent $a_i \overset{\text{approx.}}{\sim} N(0, \sigma_a^2)$, $\sigma_a^2 \geq 0$, corresponding to 'random deviations' of the assumption of identical probability differences, represent the random interaction effects, also assumed to be stochastically independent of z_{ik} , $i = 1, \dots, K$, $k = 1, 2$. Thus we have

$$\hat{\mu}_i \overset{\text{approx.}}{\sim} N(\mu, \sigma_a^2 + \sigma_i^2), \quad \sigma_i^2 = \sigma_{i1}^2 + \sigma_{i2}^2, \quad i = 1, \dots, K > 2. \quad (16)$$

An unbiased estimate of σ_{ik}^2 is given by $\hat{\sigma}_{ik}^2 = \hat{p}_{ik}(1 - \hat{p}_{ik})/(n_{ik} - 1)$, such that $\xi_i = \hat{\sigma}_{i1}^2 + \hat{\sigma}_{i2}^2$. It will be sufficient in our case to approximate $\text{var}(\hat{\sigma}_{ik}^2)$ by the delta method, thus

$$\text{var}(\hat{\sigma}_{ik}^2) \approx \left(\frac{\partial \hat{\sigma}_{ik}^2}{\partial \hat{p}_{ik}} \bigg|_{\hat{p}_{ik}=p_{ik}} \right)^2 \cdot \sigma_{ik}^2 = \left(\frac{1 - 2p_{ik}}{n_{ik} - 1} \right)^2 \cdot \frac{1}{n_{ik}} p_{ik}(1 - p_{ik}), \quad (17)$$

which by replacing p_{ik} with $\hat{p}_{ik}$ gives the estimate $\widehat{\text{var}}(\hat{\sigma}_{ik}^2)$, so that $\widehat{\text{var}}(\xi_i) = \widehat{\text{var}}(\hat{\sigma}_{i1}^2) + \widehat{\text{var}}(\hat{\sigma}_{i2}^2)$. Therefore, the procedures of section 2 and 3 can now readily be applied.

7. Examples

(a) Artificial insemination data

This example is a classical one from Snedecor and Cochran (1967, p.

290). The summary data are presented in Table V below. In research on artificial insemination of cows, a series of semen samples from a bull are sent out and tested for their viability. The summary data, $\hat{\mu}_i$, is the mean (percentages) of conceptions obtained from samples of six bulls.

The inverse variance weights $c_i = (1/\xi_i) / \left\{ \sum_{i=1}^K (1/\xi_i) \right\}$, see section 2, are $c_1 = 0.153$, $c_2 = 0.178$, $c_3 = 0.283$, $c_4 = 0.053$, $c_5 = 0.139$, $c_6 = 0.194$; so that $b_i = c_i < \frac{1}{2}$, $i = 1, \dots, 6$. That is, no single group (bull) can be said to be dominant. The between-group variance estimates are $\hat{\sigma}_{a,DL}^2 = 64.800$, $\hat{\sigma}_{a,2}^2 = 31.814$, $\hat{\sigma}_a^2(\eta) = 30.794$, and $\hat{\sigma}_a^2(\lambda_2) = 58.439$, with the 95% confidence intervals being

$$CI_1(\sigma_a^2) : [-0.450, 189.814] \doteq [0, 189.814],$$

$$\text{and } CI_2(\sigma_a^2) : [17.442, 230.142],$$

within which lie all the between-group variance estimates. We remark that the estimated degrees of freedom of the quadratic forms involved in the confidence intervals, respectively, are $\hat{\nu}_Q = 8.03$ for CI_1 and $\hat{\nu}_{Q^*} = 12.30$ for CI_2 .

Table V. Summary data on artificial insemination of cows

Bull	Sample size	Mean $\hat{\mu}_i$	Variance ξ_i
1	5	41.2	35.140
2	2	64.5	30.25
3	7	56.3	18.986
4	5	39.6	101.060
5	7	67.1	38.6395
6	9	53.2	27.716

(b) *Multiple sclerosis data*

This example is from Petitti (1994, p. 117). Table VI presents data from a meta-analysis of the effect of azathioprine treatment on progression

of disability in patients with multiple sclerosis. All studies eligible for inclusion in the meta-analysis used the Kurtzke Disability Status Scale to measure the effect of azathioprine, see Yudkin et al. (1991).

The inverse variance weights, see section 5, are $c_1 = 0.640$, $c_2 = 0.120$, $c_3 = 0.155$, $c_4 = 0.085$. We see that $c_1 > \frac{1}{2}$ so that by using procedure (15), we obtain the adjusted weights to be $b_1 = 0.484$, $b_2 = 0.173$, $b_3 = 0.221$ and $b_4 = 0.122$. That is, there is now by involving these weights no study which can be said to be dominating.

Next, the unbiased estimates of the between-group variance are $\hat{\sigma}_{a,1}^2 = -0.013$ which is truncated to give $\hat{\sigma}_{a,DL}^2 = 0.0$, and $\hat{\sigma}_{a,2}^2 = 0.011$. That is, to obtain $\hat{\sigma}_{a,DL}^2$, we have to truncate an undesirable negative value. The positive estimates suggested in this article take the values $\hat{\sigma}_a^2(\eta) = 0.021$ and $\hat{\sigma}_a^2(\lambda_2) = 0.016$. The 95% confidence intervals are $CI_1(\sigma_a^2) : [-0.028, 3.579] \doteq [0, 3.579]$, and $CI_2(\sigma_a^2) : [-0.050, 0.652] \doteq [0, 0.652]$.

Note that in this example only a few and extremely unbalanced studies are enclosed, paired with an obviously low between-study variance in relation to the (weighted) average of the within-study variances of the treatment differences. This leads to small estimates of the degrees of freedom of the quadratic forms, namely $\hat{\nu}_{Q^*} = 3.10$ used in CI_2 and $\hat{\nu}_Q = 1.70$ used in CI_1 which particularly influence the upper bounds of the confidence intervals in an unfavourable way, resulting in too conservative intervals. Here we may further remark that solving the REML equation for σ_a^2 iteratively also leads to a negative estimate (-0.0012), so that, for instance, the SAS-package gives the REML estimate 0 for σ_a^2 .

Table VI. Summary data, on the effect of azathioprine treatment on progression of disability in patients with multiple sclerosis

	Treatment			Control		
Study i	Sample size n_{i1}	Mean $\hat{\mu}_{i1}$	Variance ξ_{i1}	Sample size n_{i2}	Mean $\hat{\mu}_{i2}$	Variance ξ_{i2}
1	162	0.30	0.0098	175	0.42	0.0094
2	15	0.17	0.0540	20	0.83	0.0480
3	30	0.20	0.0403	32	0.45	0.0392
4	27	0.17	0.0705	25	0.42	0.0740

(c) Non-ulcer dyspepsia data

This example is on meta-analytic evaluation of 13 placebo-controlled clinical trials of a gastroprokinetic agent named cisapride in the treatment of non-ulcer dyspepsia. Non-ulcer dyspepsia is characterized by a variety of upper abdominal symptoms in the absence of an organic disease. Scanning the available literature yielded 13 placebo-controlled trials with cisapride which were eligible for a meta-analysis. The studies are listed by stating the first author, the year and the journal in which they were published, see Table VII.

Table VII. Data of 13 cisapride studies (Number of successes/Number of patients), estimates of the outcome measure (the risk difference) and the estimated variances

Study	Cisapride	Placebo	$\hat{\mu}_i$	ξ_i
Creytens (1984) Current Therap. Res. 36	15/16	9/16	0.375	0.020
Milo (1984) Current Therap. Res. 36	12/16	1/16	0.688	0.016
Francois (1984) Digestion 34	29/34	18/34	0.324	0.011
Derutere (1987) Clinical Therapeutics 10	42/56	31/56	0.196	0.008
Hannon (1987) Current Therap. Res. 42	14/22	6/22	0.364	0.021
Roesch (1987) Scand. J. Gastro. 22	44/54	17/55	0.506	0.007
De Nutte (1989) Clinical Therapeutics 11	14/17	7/15	0.357	0.027
Hausken (1992) Scand. J. Gastro. 27	29/58	23/58	0.103	0.007
Chung (1993) Scand. J. Gastro. 28	10/14	3/15	0.514	0.027
Van Outryve (1993) Scand. J. Gastro. 28	17/26	6/27	0.432	0.016
Al-Quorain (1995) Scand. J. Gastro. 30	38/44	12/45	0.597	0.007
Kellow (1995) Aliment. Pharm. & Therap. 9	19/29	22/30	-0.078	0.015
Yeoh (1997) J. Gastro. & Hepatol. 12	21/38	19/38	0.053	0.013

In this example, the inverse variance weights, see section 6, satisfy $c_i < \frac{1}{2}$, for all $i = 1, \dots, 13$. Therefore, in Cochran's sense, there is no single study which dominates the others. That is, we can now set $b_i = c_i$ for all i , to obtain $\hat{\sigma}_{a,DL}^2 = 0.040$, $\hat{\sigma}_{a,2}^2 = 0.044$, $\hat{\sigma}_a^2(\eta) = 0.039$ and $\hat{\sigma}_a^2(\lambda_2) = 0.036$, which are contained within the 95% confidence bounds $CI_1(\sigma_a^2) : [0.021, 0.110]$ and $CI_2(\sigma_a^2) : [0.020, 0.085]$. The corresponding estimates for the degrees of freedom are $\hat{\nu}_Q = 18.01$ and $\hat{\nu}_{Q^*} = 26.92$.

8. Conclusion

Two positive estimators of the heterogeneity variance have been derived. Their biases and standard deviations have been shown to be close to those of the DerSimonian-Laird estimator, a popular estimator of the between-study

variance in meta-analysis. The proposed estimators have relatively stable biases with respect to changes in σ_a^2 . Therefore, one is most likely not to be very far from the true parameter by using these estimators. Given the well documented difficulties in the use of iterative procedures, like ML and REML, the proposed estimators are finite and easier to implement. Another added advantage to the new estimators is that it is possible to construct explicit confidence bounds on the heterogeneity variance.

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